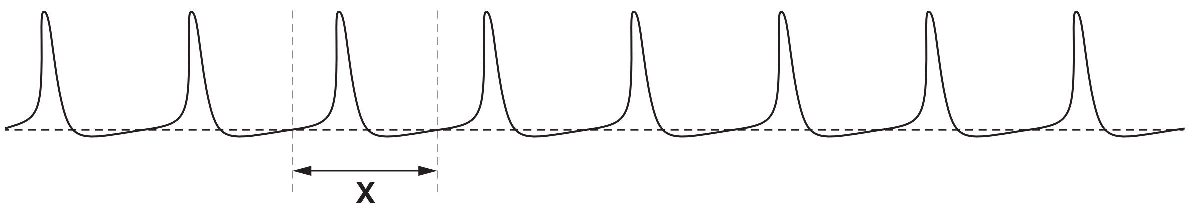
Examiner only

Answer **all** *questions.*

1. *(a)* Scientists applied different constant pressures to a sensory neurone to investigate the production of action potentials.

At the highest pressure the oscilloscope trace of the action potentials looked like this.



The time taken to complete one action potential is marked as **X** on the trace.

There were 120 action potentials produced each second.

(i) How long did each action potential (time **X**) last? Give your answer to the nearest millisecond. [2]

Time for one action potential = ... ms

At the lowest pressure applied the trace appeared as shown below.

(ii) Explain why the trace appeared this way for the lowest pressure. [2]

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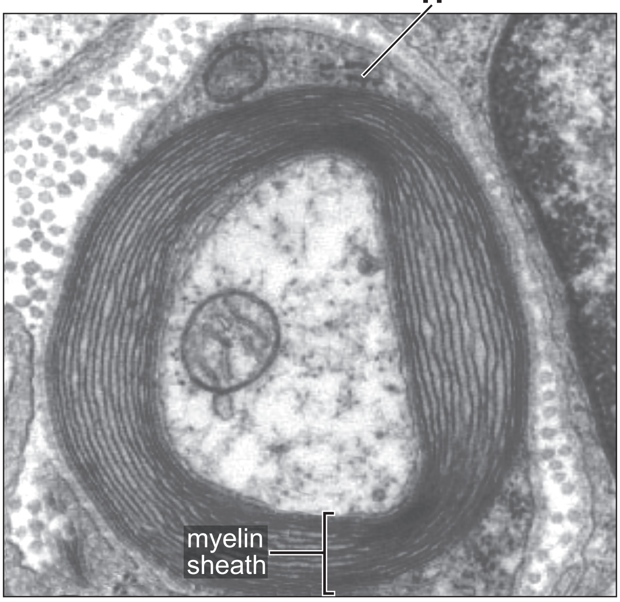
(b) The electron micrograph shows an axon surrounded by myelin sheath.



Cell **H** forms the myelin sheath.

(i) Name cell **H**. [1]

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(ii) Name the main class of biological molecule found in myelin. [1]

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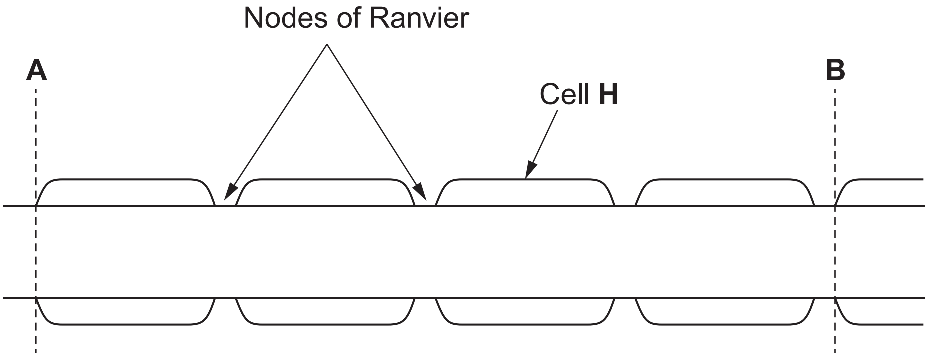
The neurone appears like this when drawn as a side section.



Drawing not to scale

The distance between **A** and **B** is 5150 µm.

The usual figure given for the speed of conduction of an action potential in a myelinated neurone is 1×10^{8} µm s^{-1}.

(iii) Calculate the time it would take an action potential to travel from point **A** to point **B** on an actual neurone. Give your answer in standard form in seconds. [3]

Time = ... s

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(c) It has been found that organophosphates, which are used in some pesticides, can cause the myelin sheath to become damaged. In people who handle organophosphates, the myelin sheath may degenerate and leave the membrane of the axon exposed.

(i) Suggest what would happen to the rate of oxygen consumption for the demyelinated neurone. Explain your answer. [4]

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(ii) Suggest **two** symptoms which could occur in someone who has demyelinated neurones. [2]

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